

YOU ALREADY OWN A GENERATOR. IT'S PARKED IN YOUR DRIVEWAY.

One box the size of a lunch pail, clamped onto your car battery, and the fridge light comes back on. Under 200 dollars on top of the car you already own.

The power is out and your fridge just started a four hour clock. That is how long the Department of Agriculture says a closed fridge keeps food safe once it goes dark. A stock car battery, engine off, runs an average fridge for roughly 5 to 8 hours on paper. Idle the engine for an hour or two every few hours, outside and never in the garage, and that fridge stays cold as long as the gas lasts.

THE HONEST NUMBERS

100 to 200 W	What the compressor pulls while it runs. It runs 30 to 50 percent of the day, so a typical fridge burns about 1,500 watt hours a day, call it 60 some watts around the clock.
70 amp hours	A common group 24 car battery. A decent inverter hands you about 87 cents of every dollar, so the fridge pulls about 6 amps out of the battery, hour after hour.
Half	The working rule. A starting battery was never built to be drained. Budget 35 of the 70 amp hours. 35 divided by 6 is a little under 6 hours. A group 27 gets you closer to 7.
30 to 50 amps	What the alternator makes at idle. The fridge wants 6. The engine runs the fridge outright and refills the battery with what is left.
1 to 2 dollars	An hour of idling at about four dollars a gallon. Six or eight hours a day keeps the fridge alive for 4 to 16 dollars. Set that against a week of groceries.



Car outside, inverter on a dry crate, one cord to one appliance. The fridge is the first thing you save.

WATCH THE METER, NOT THE CLOCK

Shut the inverter off and let the battery rest a few minutes, because the voltage only tells the truth at rest. About 12.6 volts is full. Down around 12.1 or 12.2 you are near half, so start the engine. A lead acid battery drinks fast when it is low and slow when it is nearly full, so it will not be brimming when you shut the engine off. Run it to zero two or three times and you have a battery that cannot crank a cold engine, in a blackout, with warm meat. This is for an ordinary gas car with a 12 volt battery under the hood. Hybrids and electric cars play by different rules.

INVERTER, CABLES, FUSE, CORD

Run the whole setup on a normal Saturday. Some newer cars shut themselves off at idle; find out then, not at midnight in a blackout.

140 to 250 dollars

Pure sine wave inverter, 1,000 to 1,500 watts continuous, surge rating at least double. An old compressor grabs three to seven times its running load for a second, a thousand watts or more. A cheap inverter sees that and shuts off. Modified sine runs a compressor hot and can cook it.

25 to 45 dollars

Two short battery cables, 4 gauge copper as a floor, heavier for a bigger inverter or a longer run. Six amps average turns into a hundred amps during the spike, and thin cable cooks.

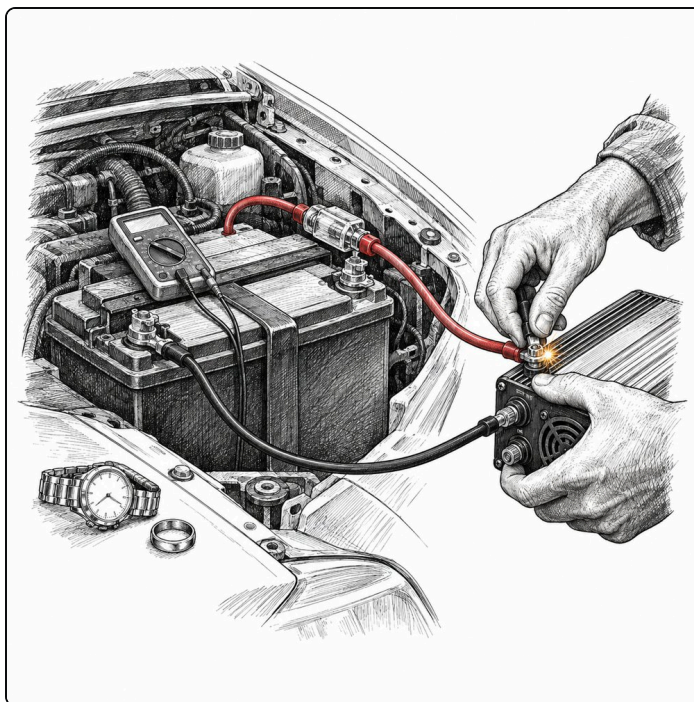
about 20 dollars

Fuse and holder on the positive cable, as close to the battery as the inverter manual says, usually 150 to 200 amps, cable rated for it too. Without it, one dropped wrench turns that cable into a welding rod.

HOOKUP, IN ORDER

- 1 Rings and watches off.** Engine off. Inverter off. Hands and metal tools never across both posts at once.
- 2 Bolt the cables to the inverter first.**
- 3 Red to the battery's positive post**, fuse in line, close to the battery.
- 4 Black to the negative post.**
- 5 Make the final connection at the inverter end**, away from the battery, or through a battery switch. The last connection always throws a spark and a lead acid battery breathes hydrogen. Never make it at the battery. No smoking, no flame, ever.
- 6 Extension cord:** 12 gauge outdoor, as short as you can manage. A long skinny cord drops enough voltage to trip the shutdown.
- 7 Flip the inverter on**, plug in the fridge, listen for the compressor to catch.
- 8 Keep the inverter off wet ground** and out of the rain.
- 9 Never plug the inverter into a wall outlet or a dryer plug** to feed the house. That sends power back up the line and can kill the crew fixing it. One cord, one appliance.

Want the real number for your fridge? A plug in watt meter costs about the price of lunch. Leave it on the fridge for a day.



Black on first. The last connection happens at the inverter end, never at the battery.

FREE MOVES THAT STRETCH THE CLOCK

- ☐ Keep the door shut. A full freezer holds about 48 hours with no power, 24 if half full. Every peek spends that clock.
- ☐ Fill empty freezer space with jugs of water today. Frozen jugs are thermal mass; when the power drops they slide into the fridge and a cooler and keep working.
- ☐ A separate chest freezer stays shut and gets a turn on the inverter only when it needs one.
- ☐ A cheap fridge thermometer takes the guessing out. Food above 40 degrees for more than 2 hours goes in the trash. Do not taste it to find out.
- ☐ For a four hour outage you need a closed door, not a rig. The rig is for the fourth hour and the fourth day.

THE LINES YOU DO NOT CROSS, AND THE UPGRADE

OUTSIDE, 20 FEET, ALARMS ON

This is the part that puts people in the hospital.

LINES YOU DO NOT CROSS

- ☐ Never idle the car inside an attached garage, not even with the door up. Carbon monoxide has no smell, spreads through the whole house, and kills people every single winter.
- ☐ The car sits outside, at least 20 feet from the house, tailpipe pointed away from every window and door. Parked, parking brake set.
- ☐ Bring the cord in on the far side of the house from the tailpipe. Do not let the door pinch it. A towel in the gap will not keep carbon monoxide out.
- ☐ Battery powered carbon monoxide alarm on every level and outside every bedroom. Test them before you turn the key.
- ☐ No live connection at the battery. No smoking or flame near it. Lead acid breathes hydrogen and lives outdoors or in an open garage.
- ☐ Never backfeed the house through a wall outlet or dryer plug.
- ☐ Lead acid loses a real chunk of capacity when it is freezing, so your hours shrink on a January night. Lithium cells are damaged if charged below freezing; buy one with low temperature charge protection and keep it inside where it is warm.

This one is for education. Know your own car, your own skills, and your local rules before you clamp anything on.



Garage idling fills the bedrooms. Twenty feet out, tailpipe away, is the only spot for the car.

THE UPGRADE, WHEN ONE NIGHT IS NOT ENOUGH

**130 to 250
dollars**

100 amp hour marine deep cycle. Still lead acid, same half budget, 50 usable amp hours, around 8 hours of fridge. Breathes hydrogen while charging, so outdoors or open garage.

**180 to 280
dollars**

100 amp hour lithium iron phosphate. About 90 usable amp hours, about 15 hours of fridge per charge, engine off all night. A third the weight, no hydrogen, can live indoors near the fridge with the terminals covered. Inverter beside it, not on it.

**100 to 150
dollars**

Battery to battery charger. Never clamp lithium straight onto the car; it can cook the alternator. The charger limits the flow to 20 or 30 amps. Cable sized for the whole run, a fuse at both batteries.

**500 to 700
dollars**

All in with the lithium battery and charger. Against a 2,000 watt inverter generator at 340 on sale, over 700 for the name brands, plus its own gas, oil, and spot outside. A 512 watt hour power station, 255 to 469 dollars, is about a third of a day of fridge.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next blackout does not catch you flat footed.